AI Task Platform – Architecture Document

1. Overview

The AI Task Platform is a distributed system deployed using Kubernetes. It consists of multiple microservices including frontend, backend, database, cache, and worker services.

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2. System Components

Frontend

- Built using React
- Served via Nginx
- Exposed using NodePort

Backend

- Node.js API server
- Handles business logic and API requests

MongoDB

- Stores persistent data

Redis

- Used for caching and task queue

Worker

- Processes background tasks asynchronously

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3. Architecture Diagram (describe if not drawing)

User → Frontend → Backend → (MongoDB + Redis) → Worker

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4. Deployment Architecture

- Docker used for containerization
- Kubernetes used for orchestration
- Each service deployed using Deployment + Service

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5. Service Communication

Service	Communicates With
Frontend	Backend
Backend	MongoDB, Redis
Worker	MongoDB, Redis

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6. Challenges & Fixes

- Invalid image name → corrected Docker image
- CrashLoopBackOff → fixed nginx config
- Service communication → used Kubernetes service names

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7. Scalability

- Horizontal scaling via Kubernetes replicas
- Worker supports parallel task execution

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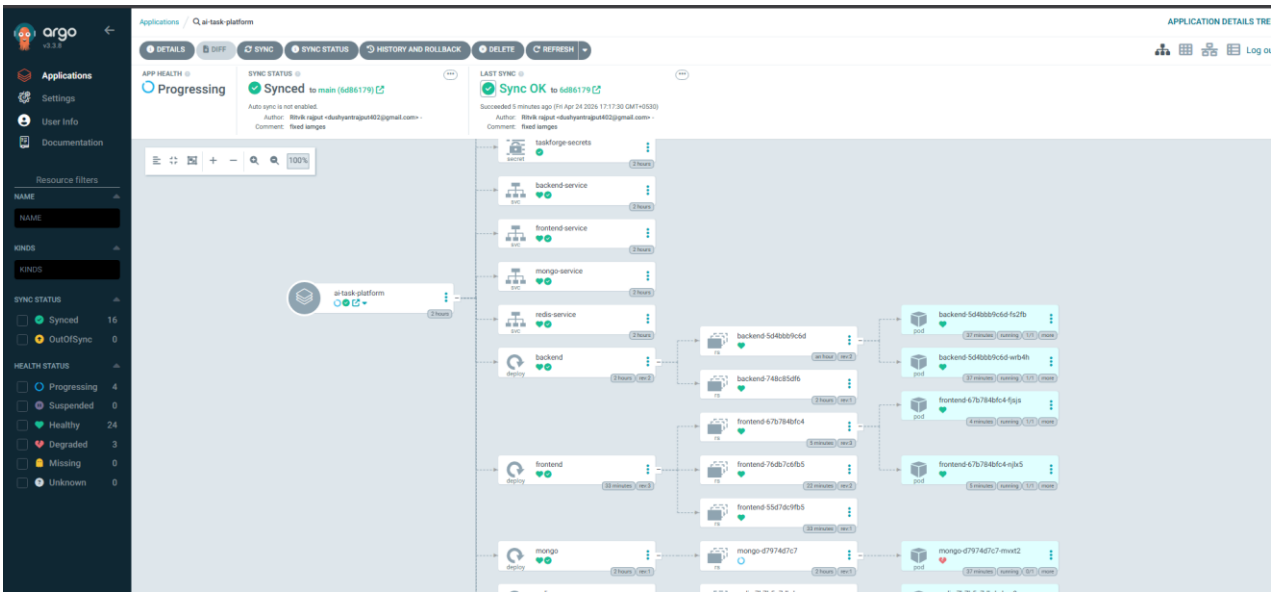
8. Conclusion

The system demonstrates microservices deployment using Kubernetes with proper service communication and scalability.